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(Yapa Mudiyanselage Anuradha Ishani Yapa)

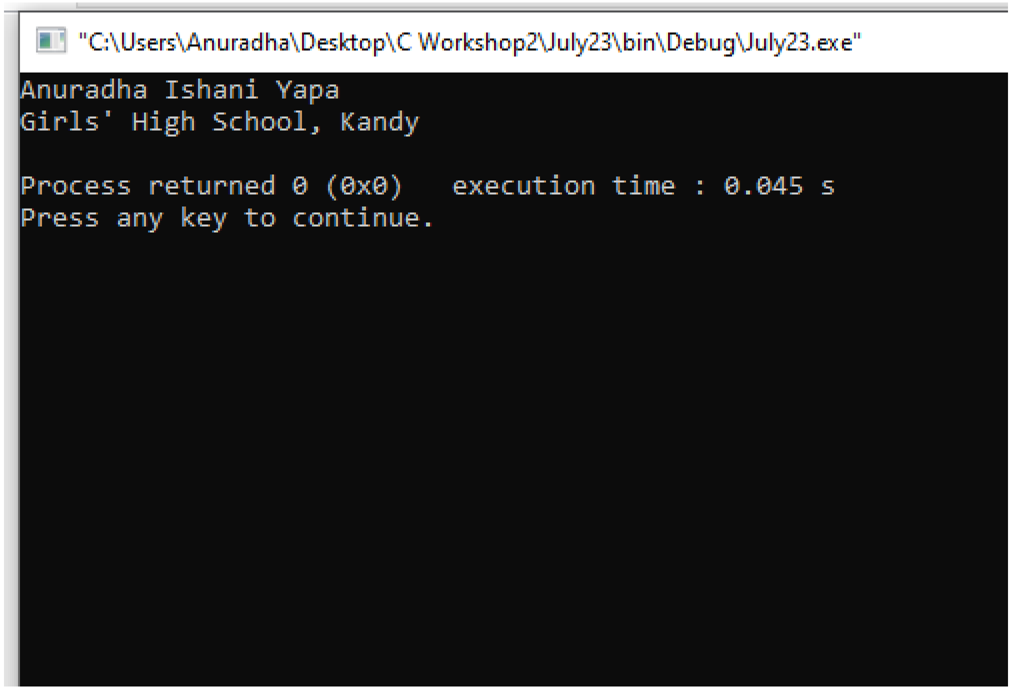
Index Number:
20238

Practical 01

1)

main.c

```c
#include <stdio.h>
int main()
{
    printf("Anuradha Ishani Yapa\n");
    /* printf() is a built-in function of the
       Standard Input Output Header File, used to
       print character, string, integer, float, octal
       and hexadecimal values onto the
       output screen */

    // \n is an escape sequence that denotes a newline character

    printf("Girls' High School, Kandy\n");


}

```


"C:\Users\Anuradha\Desktop\C Workshop2\July23\bin\Debug\July23.exe"
Anuradha Ishani Yapa
Girls' High School, Kandy
Process returned 0 (0x0)   execution time : 0.045 s
Press any key to continue.

2)

main.c

```c
#include <stdio.h>
int main()
{
    printf("*\n\n");
    printf("**\n\n");
    printf("***\n\n");
    printf("****\n\n");
    printf("*****\n\n");


    /*Semicolon ends statements in C Programming Language.
    Semicolon states that the statement to declare the add
    function has been terminated, and whatever follows is
    a new statement

    \n is being used twice in each printf statement which
    creates double line breaks */
}
```

"C:\Users\Anuradha\Desktop\C Workshop2\July23\bin\Debug\July23.exe"

```
*

**

***

****

*****


Process returned 0 (0x0)   execution time : 0.040 s
Press any key to continue.
```

3)

main.c

```c
#include <stdio.h>
int main()
{
    int x;
    printf("Enter an integer value: ");
    scanf("%d",&x);

    /* Variable is a temporary memory location in RAM,
       used to store and retrieve a data type.
       There are different types of variables such as
       integer, float, double, char, string. */

    float y;
    printf("Enter a float value: ");
    scanf("%f",&y);

    double z;
    printf("Enter a double value: ");
    scanf("%lf",&z);

    char c;
    printf("Enter your favorite letter: ");
    scanf(" %c",&c);

```

```c

    /* scanf() function is used for entering input data.
    This function can take all types of input values
    (numeric, character, string).
    It has at least two parameters. First parameter is a
    control string, containing the conversion specifier
    within double quotes.
    The other parameter is the variable address.

    Conversion specifiers of some data types are listed following.

    integer - %d
    float   - %f
    double  - %f
    long double - %lf
    a single character - %c
    a string - %s

    */

    printf("\nInteger value: %d\n",x);
    printf("Float value: %.3f\n",y);
    printf("Double value: %lf\n",z);
    printf("Your favorite letter: %c\n",c);
}

```

"C:\Users\Anuradha\Desktop\C Workshop2\July23\bin\Debug\July23.exe"

```
Enter an integer value: 5
Enter a float value: 4.3
Enter a double value: 343.222443
Enter your favorite letter: A

Integer value: 5
Float value: 4.300
Double value: 343.222443
Your favorite letter: A

Process returned 0 (0x0)   execution time : 11.013 s
Press any key to continue.
```

4)

*main.c

```c
#include <stdio.h>
int main()
{
    int n1,n2;
    //two integer type variables are declared

    printf("Enter the first integer: ");
    scanf("%d",&n1);
    //user inputs an integer value for the variable n1

    printf("Enter the second integer: ");
    scanf("%d",&n2);
    //user inputs an integer value for the variable n2

    int total;
    total=n1+n2;
    /*an integer type variable named "total" is declared
      and the sum of n1 and n2 is calculated and stored in there */

    printf("\nTotal:\n");
    printf("%d + %d = %d\n",n1,n2,total);
    //the sum of the two user-input values are displayed in the console window
}

```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the first integer: 5
Enter the second integer: 7

Total:
5 + 7 = 12

Process returned 0 (0x0)   execution time : 7.141 s
Press any key to continue.
```

5)

main.c

```c
#include <stdio.h>
int main()
{
    float n1,n2;
    //two variables of type float are declared

    printf("Enter a number with decimals: ");
    scanf("%f",&n1);
    //user-input floating-point number is stored in n1

    printf("Enter another number with decimals: ");
    scanf("%f",&n2);
    //user-input floating-point number is stored in n2

    float tot=n1+n2;
    //sum of the two values are calculated and stored in tot
    float avg=tot/2;
    //average is calculated and stored in avg

    printf("\nAverage: %.2f\n",avg);
    //average is displayed/printed on the console window
}

```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter a number with decimals: 5.4
Enter another number with decimals: 5.4

Average: 5.40

Process returned 0 (0x0)   execution time : 7.327 s
Press any key to continue.
```

6)

*main.c

```c
#include <stdio.h>
int main()
{
    char name[50];
    printf("Enter the Student Name: ");
    scanf("%s",&name);

 /* C programming language does not have a specific "String"
    data type, the way some other programming languages such
    as C++ and Java do.

    Instead, C stores strings of characters as arrays of chars.

    The length to print or terminate the string must be specified. */

    int byear;
    printf("Enter the Birth Year: ");
    scanf("%d",&byear);

    int age=2020-byear;

    printf("\nStudent Name: %s",name);
    //%s will be replaced with the name that the user typed
    printf("\nAge: %d\n",age);
    //%d will be replaced with the calculated age
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the Student Name: John
Enter the Birth Year: 1999

Student Name: John
Age: 21

Process returned 0 (0x0)   execution time : 32.080 s
Press any key to continue.
```

7)

*main.c

```c
#include <stdio.h>
int main()
{
    int x;
    printf("Enter a value for x: ");
    scanf("%d",&x);

    int y;
    printf("Enter a value for y: ");
    scanf("%d",&y);

    printf("\nBefore Swap\n");
    printf("Value in x: %d\n",x);
    printf("Value in y: %d\n",y);

    int swap=x;
    //in order to swap the two numbers, a third variable is declared.
    //the value in x is assigned to the newly-declared variable.
    x=y;
    //the value in y is assigned to x.
    y=swap;
    //finally the value in swap(or initially x) is assigned to y.
    //now the two numbers have been swapped

```

```c
    printf("\nAfter Swap\n");
    printf("Value in x: %d\n",x);
    printf("Value in y: %d\n",y);
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter a value for x: 10
Enter a value for y: 20

Before Swap
Value in x: 10
Value in y: 20

After Swap
Value in x: 20
Value in y: 10

Process returned 0 (0x0)   execution time : 6.220 s
Press any key to continue.
```

8)

main.c

```c
#include <stdio.h>
int main()
{
    printf("The color: %s\n","blue");
    //in C, double quotes are used to identify string literals
    //the word "blue" will be printed in the place of %s
    printf("First number: %d\n",12345);
    //the specifier for the integer data type is %d
    printf("Second number: %04d\n",25);
    //25 will be displayed in 4 digits (0025) due to the use of %04d
    printf("Third number: %i\n",1234);
    //another specifier for integer data type is %i
    printf("Float number: %3.2f\n",3.14159);
    //the decimal number(3.14159) will be rounded to two decimal places
    //%.2f can be written instead of %3.2f
    printf("Hexadecimal: %x\n",255);
    //the hexadecimal equivalent of the decimal number, 225 will be printed
    //in the place of %x
    printf("Octal: %o\n",255);
    //the octal equivalent of the decimal number, 225 will be printed
    //in the place of %o
    printf("Unsigned value: %u\n",150);
    //the unsigned value of 150 will be displayed
    printf("Just print the percentage sign: %%\n",10);
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
The color: blue
First number: 12345
Second number: 0025
Third number: 1234
Float number: 3.14
Hexadecimal: ff
Octal: 377
Unsigned value: 150
Just print the percentage sign: %

Process returned 0 (0x0)   execution time : 2.460 s
Press any key to continue.
```

Practical 02

1)

```c
#include <stdio.h>
int main()
{
    int age;
    //an integer type varibale is declared
    printf("HI, HOW OLD ARE YOU? ");
    //the user is asked to enter his/her age
    scanf("%d",&age);
    //scanf() function reads formatted data from stdin
    //stdin is usually the keyboard, unless redirected

    printf("\n\n");
    //the computer skips two lines

    printf("WELCOME %d\n",age);
    printf("LET'S BE FRIENDS!\n");
    //printf() function prints above statements onto the output screen
}

```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
HI, HOW OLD ARE YOU? 26


WELCOME 26
LET'S BE FRIENDS!

Process returned 0 (0x0)   execution time : 10.045 s
Press any key to continue.
```

2)

main.c

```c
#include <stdio.h>
int main()
{
    printf("%8d %8d %8d\n",2,4,8);
    //%d prints an integer value
    //%8d is the Right Align. It keeps 8 spaces
    //before printing the integer

    printf("%8d %8d %8d\n",3,9,27);
    printf("%8d %8d %8d\n",4,16,64);
    printf("%8d %8d %8d\n",5,25,125);

    printf("\n\n\n");
    printf("\t2 \t4 \t8\n");
    //\t is used for tab space
    //\t keeps a tab space before printing the integer

    printf("\n\n\n");
    printf("%-8d %-8d %-8d\n",2,4,8);
    //%-8d is the Left Align. It keeps 8 spaces
    //after printing the integer
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
       2        4        8
       3        9       27
       4       16       64
       5       25      125



        2       4       8



2        4        8

Process returned 0 (0x0)   execution time : 4.062 s
Press any key to continue.
```

3)

*main.c

```c
#include <stdio.h>
int main()
{
    int d;
    printf("Enter the distance travelled(m): ");
    scanf("%d",&d);

    int t;
    printf("Enter the time taken(s): ");
    scanf("%d",&t);

    printf("\nAverage Speed: %d m/s\n",d/t);

 /* If integer variables are used to store distnace and time,
    and if decimal values are entered into those integer
    varibales, the program doesn't run as expected.

    Integer varibales can't store decimal values. Therefore
    the program crashes after a decimal value is entered
    for distance (the program doesn't let the user to
                  input the time taken).
    This problem can be solved by using floating-point variables */
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the distance travelled(m): 7.77
Enter the time taken(s):
Average Speed: 0 m/s

Process returned 0 (0x0)   execution time : 77.483 s
Press any key to continue.
```

main.c

```c
#include <stdio.h>
int main()
{
    float d;
    printf("Enter the distance travelled(m): ");
    scanf("%f",&d);

    float t;
    printf("Enter the time taken(s): ");
    scanf("%f",&t);

    //to input decimal numbers, floating-point variables
    //must be used.

    printf("\nAverage Speed: %.2f m/s\n",d/t);
    //Average speed is rounded upto two decimal palces and displayed

}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the distance travelled(m): 67.55
Enter the time taken(s): 65.55

Average Speed: 1.03 m/s

Process returned 0 (0x0)   execution time : 13.669 s
Press any key to continue.
```

4)

main.c

```c
#include <stdio.h>
int main()
{
    float f;
    printf("Enter the temperature in Fahrenheit: ");
    scanf("%f",&f);

    float c;
    c=(f-32)*5/9;

    printf("Temperature in Celsius: %.2f\n",c);
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the temperature in Fahrenheit: 68
Temperature in Celsius: 20.00

Process returned 0 (0x0)   execution time : 2.425 s
Press any key to continue.
```

```c
#include <stdio.h>
int main()
{
    float f;
    printf("Enter the temperature in Fahrenheit: ");
    scanf("%f",&f);

    float c;
    c=(f-32)*5/9;

    printf("Temperature in Celsius: %.2f\n",c);
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the temperature in Fahrenheit: 150
Temperature in Celsius: 65.56

Process returned 0 (0x0)   execution time : 2.676 s
Press any key to continue.
```

main.c

```c
#include <stdio.h>
int main()
{
    float f;
    printf("Enter the temperature in Fahrenheit: ");
    scanf("%f",&f);

    float c;
    c=(f-32)*5/9;

    printf("Temperature in Celsius: %.2f\n",c);
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the temperature in Fahrenheit: 212
Temperature in Celsius: 100.00

Process returned 0 (0x0)   execution time : 2.017 s
Press any key to continue.
```

main.c

```c
#include <stdio.h>
int main()
{
    float f;
    printf("Enter the temperature in Fahrenheit: ");
    scanf("%f",&f);

    float c;
    c=(f-32)*5/9;

    printf("Temperature in Celsius: %.2f\n",c);
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the temperature in Fahrenheit: 0
Temperature in Celsius: -17.78

Process returned 0 (0x0)   execution time : 1.917 s
Press any key to continue.
```

main.c

```c
#include <stdio.h>
int main()
{
    float f;
    printf("Enter the temperature in Fahrenheit: ");
    scanf("%f",&f);

    float c;
    c=(f-32)*5/9;

    printf("Temperature in Celsius: %.2f\n",c);
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the temperature in Fahrenheit: -22
Temperature in Celsius: -30.00

Process returned 0 (0x0)   execution time : 2.176 s
Press any key to continue.
```

main.c

```c
#include <stdio.h>
int main()
{
    float f;
    printf("Enter the temperature in Fahrenheit: ");
    scanf("%f",&f);

    float c;
    c=(f-32)*5/9;

    printf("Temperature in Celsius: %.2f\n",c);
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the temperature in Fahrenheit: -100
Temperature in Celsius: -73.33

Process returned 0 (0x0)   execution time : 2.181 s
Press any key to continue.
```

main.c

```c
#include <stdio.h>
int main()
{
    float f;
    printf("Enter the temperature in Fahrenheit: ");
    scanf("%f",&f);
    //temperature in Fahrenheit is stored in f

    float c;
    c=(f-32)*5/9;
    //Fahrenheit temperature is converted to Celsius and stored in c

    printf("Temperature in Celsius: %.2f\n",c);
    //temperature in Celsius is displayed
}
```

5)

main.c

```c
1  #include <stdio.h>
2  int main()
3  {
4      int i=5,j=6,k;
5      k=++i + j++;
6
7      /*first increment the value of i by 1 and then
8      add that value to j and finally assign the addition to k.
9
10     After the program executes the line 5,
11     the value of j will be 7.
12
13     i - pre-increment
14     j - post-increment */
15
16
17     printf("%d %d %d",i,j,k);
18
19     /*The output of the above program will be
20
21     i=6
22     j=7
23     k=12 */
24 }
25
```

6)

main.c

```
1  #include <stdio.h>
2  int main()
3  {
4      int i=1,j;
5      j= 2 + 2 * i++;
6
7      /*First multiply i value(1) by 2 and then add 2 to the result.
8      After that, assign the final result to j.
9
10     After the program executes the 5th line,
11     the value of i will be 2.
12
13     i - post-increment */
14
15
16     printf("%d %d",i,j);
17
18     /*The output of the above program will be
19
20     i=2
21     j=4 */
22 }
23
```

7)

*main.c

```c
#include <stdio.h>
int main()
{
    int a=2,b=7,c=10;
    c = a==b;

    printf("%d",c);

    /*The output of the above program will be 0 (c=0).

    The assignment will set c to 0.

    The == equal operator yields 1 if its operands are
    equal and 0 if the operands are not equal.

    In the above instance, the operands are not equal.
    Therefore, c will be 0. */

}
```

8)

*main.c

```c
#include <stdio.h>
int main()
{
    int a=22;

    if(a=0){
        printf("true");
    }

    else{
        printf("false");
    }

    //the output of the above program will be false

    //"=" is an assignment operator
    //so 0 will be assigned to 'a variable'
    //and therefore the condition will be false
}
```

Practical 03

1)

main.c

```c
#include <stdio.h>
int main()
{
    int n1,n2,max;
    printf("Enter the first number: ");
    scanf("%d",&n1);
    //user inputs a value for n1

    printf("Enter the second number: ");
    scanf("%d",&n2);
    //user inputs a value for n2

    if(n1>n2)
        max=n1;
    //if the condition is true, n1 value will be assigned to max
    else
        max=n2;
    //if the condition is false, n2 value will be assigned to max

    printf("\nHighest Value: %d\n",max);
    //Highest value will be printed on the console
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

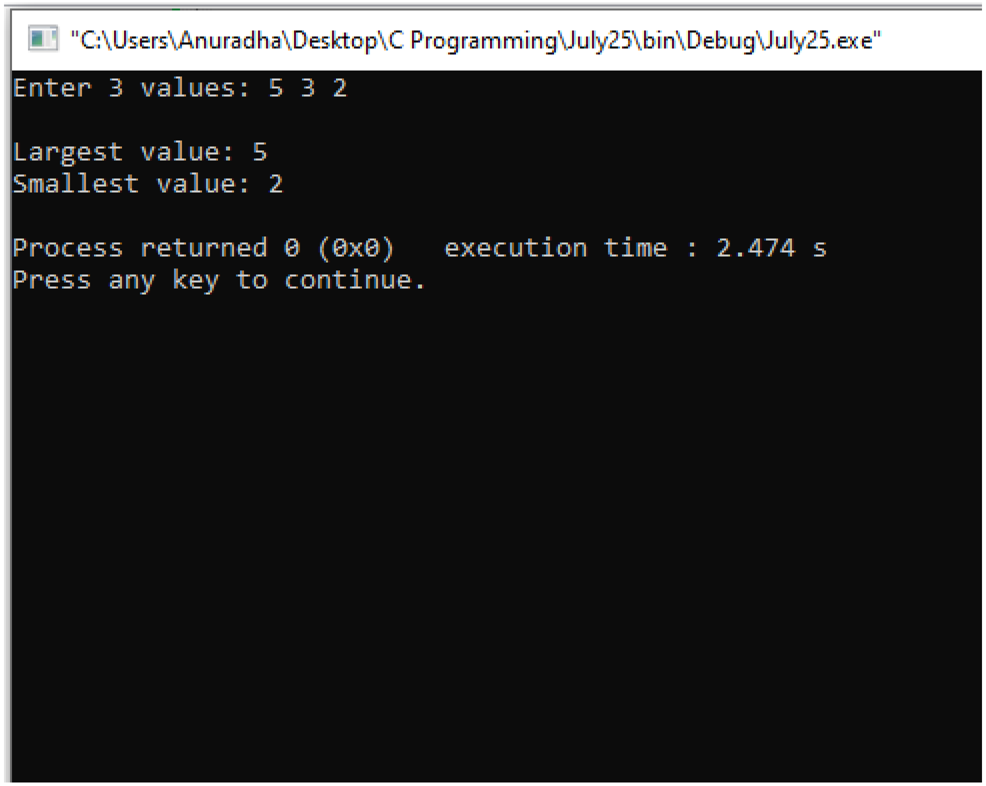
```
Enter the first number: 6
Enter the second number: 8

Highest Value: 8

Process returned 0 (0x0)   execution time : 3.049 s
Press any key to continue.
```

2)

main.c

```c
#include <stdio.h>
int main()
{
    int n1,n2,n3,max,min;
    printf("Enter 3 values: ");
    scanf("%d %d %d",&n1,&n2,&n3);
    //scanf() function takes 3 integer values to the program

    max=n1;
    //the first value that the user inputs is assigned to max

    if(n2>max)
        max=n2;
    //if the second value is greater than the first one,
    //second value will be assigned to max
    if(n3>max)
        max=n3;
    //if the third value is greater than the second one,
    //third value will be assigned to max

    printf("\nLargest value: %d\n",max);
    //Largest value is displayed

```

```c
    min=n1;
    //the first value that the user inputs is assigned to min

    if(n2<min)
        min=n2;
    //if the second value is smaller than the first one,
    //second value will be assigned to min
    if(n3<min)
        min=n3;
    //if the third value is smaller than the second one,
    //third value will be assigned to min

    printf("Smallest value: %d\n",min);
    //Smallest value is displayed
}
```


"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"
Enter 3 values: 5 3 2
Largest value: 5
Smallest value: 2
Process returned 0 (0x0)   execution time : 2.474 s
Press any key to continue.

3)

main.c

```c
#include <stdio.h>
int main()
{
    char name[50];
    printf("Enter Employee Name: ");
    scanf("%s",&name);

    float bs;
    printf("Enter Basic Salary: Rs.");
    scanf("%f",&bs);

    float ns;
    if(bs<5000)
        ns=bs*(1.05);
    //if the employee's basic salary is less than Rs.5000,
    //an increment of 5% is added to the basic salary
    //and the new salary is calculated
    if(bs>=5000 && bs<10000)
        ns=bs*1.1;
    /*if the employee's basic salary is greater than or equal to
      Rs.5000 AND less than Rs.10000,an increment of 10% is added
      to the basic salary and the new salary is calculated */
    if(bs>=10000)
        ns=bs*1.15;
    /*if the employee's basic salary is greater than or equal to
      Rs.10000, an increment of 15% is added to the basic salary
      and the new salary is calculated */
```

```c

    printf("\nEmployee Name: %s",name);
    //Employee Name is displayed
    printf("\nNew Salary: Rs.%.2f\n",ns);
    //Employee's new salary is displayed
}

```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter Employee Name: Aruna
Enter Basic Salary: Rs.10000

Employee Name: Aruna
New Salary: Rs.11500.00

Process returned 0 (0x0)   execution time : 7.338 s
Press any key to continue.
```

4)

*main.c

```c
#include <stdio.h>
int main()
{
    float r;

    printf("Enter the radius of the circle(m): ");
    scanf("%f",&r);
    //program reads in the radius of the circle

    printf("\n");
    //line break
    printf("Diameter of the circle: %.2f\n",r*2);
    //Diameter is calculated and printed
    printf("Circumference of the circle: %.2f\n",2*3.14*r);
    //Circumference is calculated and printed
    printf("Area of the circle: %.2f\n",3.14*(r*r));
    //Area is calculated and printed


    //calculations are performed inside the printf statements
    //conversion specifier %f is used due to the use of float variables

}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the radius of the circle(m): 7

Diameter of the circle: 14.00
Circumference of the circle: 43.96
Area of the circle: 153.86

Process returned 0 (0x0)   execution time : 3.808 s
Press any key to continue.
```

5)

*main.c

```c
#include <stdio.h>
int main()
{
    int n1,n2;
    printf("Enter two numbers: ");
    scanf("%d %d",&n1,&n2);
    //program reads in two integer values

    if(n1%n2==0)
        printf("\n%d is a multiple of %d.\n",n1,n2);
    //if the modulus is equal to 0, the first integer
    //is a multiple of the second

    else
        printf("\n%d is not a multiple of %d.\n",n1,n2);
    //if the modulus is not equal to 0, the first integer
    //is not a multiple of the second
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter two numbers: 18 3

18 is a multiple of 3.

Process returned 0 (0x0)   execution time : 3.889 s
Press any key to continue.
```

6)

main.c

```c
#include <stdio.h>
int main()
{
    printf("A  B  C  a  b  c  0  1  2  $  *  +  /\n");
    printf("%d %d %d %d %d %d %d %d %d %d %d %d %d\n",
           'A','B','C','a','b','c','0','1','2','$','*','+','/');

    /*Integer equivalents of the above uppercase letters,
    lowercase letters, digits and special symbols are
    printed. */
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
A  B  C  a  b  c  0  1  2  $  *  +  /
65 66 67 97 98 99 48 49 50 36 42 43 47

Process returned 0 (0x0)   execution time : 1.243 s
Press any key to continue.
```

7)

main.c

```c
#include <stdio.h>
int main()
{
    char name[30];
    printf("Enter Salesman's name: ");
    scanf("%s",&name);
    //program reads in the Salesman's name
    //conversion specifier '%s' is used, as a string value is read in

    float bs;
    printf("Enter Salesman's Basic Salary: Rs.");
    scanf("%f",&bs);
    //program reads in the Salesman's basic salary

    int y;
    printf("Enter the number of years of service of the Salesman: ");
    scanf("%d",&y);
    //program reads in the number of years
    //conversion specifier '%d' is used, as an integer value is read in

    float ts=bs;
    //at the initial stage, basic salary is assigned to total sales

    if(y>5)
        ts=bs*1.1;
    //10% additional allowance of basic salary is granted to salesmen with
    //over 5 years of service

```

```c

    char loc;
    printf("Type 'C' if the Salesman works in Colombo, if otherwise, type 'N': ");
    scanf(" %c",&loc);

    if(loc=='C')
        ts=ts+2500;
    //Rs.2500/= is added to the new salary if the salesman is located in Colombo

    float ms;
    printf("Enter the monthly sales of the Salesman: Rs.");
    scanf("%f",&ms);

    if(ms>=0 && ms<25000)
        ts=ts+(ms*0.1);
    if(ms>25000 && ms<50000)
        ts=ts+(ms*0.12);
    if(ms>=50000)
        ts=ts+(ms*0.15);
    //monthly bonus payment of the salesman is calculated according to
    //each employee's monthly sales amount (in Rupees)

    printf("\nGross Monthly Remuneration of %s: Rs.%.2f\n",name,ts);
    //gross monthly remuneration of the salesman is printed
}

```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter Salesman's name: Nimal
Enter Salesman's Basic Salary: Rs.12000
Enter the number of years of service of the Salesman: 7
Type 'C' if the Salesman works in Colombo, if otherwise, type 'N': C
Enter the monthly sales of the Salesman: Rs.30000

Gross Monthly Remuneration of Nimal: Rs.19300.00

Process returned 0 (0x0)   execution time : 33.630 s
Press any key to continue.
```

Practical 04

Part A

1)

main.c

```c
#include <stdio.h>
int main()
{
    int num;
    printf("Enter a number: ");
    scanf("%d",&num);
    //program reads in the user-input number

    if(num!=0){
    if(num%2==0)
        printf("\n%d is even.\n",num);
    //if the remainder is equal to 0, when the number
    //is divided by 2, that number is even

    else
        printf("\n%d is odd.\n",num);}
    //if the remainder is not equal to 0, when the number
    //is divided by 2, that number is odd
}

```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter a number: 7

7 is odd.

Process returned 0 (0x0)   execution time : 6.252 s
Press any key to continue.
```

*main.c

```c
#include <stdio.h>
int main()
{
    int num;
    printf("Enter a number: ");
    scanf("%d",&num);
    //scanf accepts the user-input from standard in

    int x;
    //an integer type variable is declared

    if(num!=0)
    {
        if(num%2==0)
            x=1;
        //if the modulus is 0, 1 is assigned to x
        else
            x=2;
        //if the modulus is not 0, 2 is assigned to x

    }
    //the user-input will be checked only if it is not equal to 0

    printf("\n");
    //line break
```

```c
    switch(x)
    {
        case 1 : printf("%d is an even number.\n",num); break;
        //if x is 0, the user-input is an even value
        case 2 : printf("%d is an odd number.\n",num);
        //if x is not 0, the user-input is an odd value
    }
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter a number: 10

10 is an even number.

Process returned 0 (0x0)   execution time : 0.910 s
Press any key to continue.
```

2)

```c
#include <stdio.h>
int main()
{
    float n1;
    printf("Enter the first number: ");
    scanf("%f",&n1);
    //the program reads in the first user-input

    float n2;
    printf("Enter the second number: ");
    scanf("%f",&n2);
    //the program reads in the second user-input

    char o;
    printf("\nEnter \n  + for Addition \n  - for Subtraction"
           "\n  * for Multiplication \n  / for Division : ");
    scanf(" %c",&o);
    //program displays a menu and the user selects the desired operator

    printf("\n");
    //line break
```

```c
    switch(o)
    {
        case '+' :printf("Addition:\n%.3f + %.3f = %.3f\n",n1,n2,n1+n2); break;
        case '-' :printf("Subtraction:\n%.3f - %.3f = %.3f\n",n1,n2,n1-n2); break;
        case '*' :printf("Multiplication:\n%.3f * %.3f = %.3f\n",n1,n2,n1*n2); break;
        case '/' :printf("Division:\n%.3f / %.3f = %.3f\n",n1,n2,n1/n2); break;
        default :printf("Invalid Operator\n");
    }

    /* Switch statements allow a variable to be tested for equality against a list of
    values. Each value is called a case, and the variable being switched on is
    checked for each switch case.

    There can be any number of case statements within a switch. Each case is followed
    by the value to be compared to and a colon.

    When a break statement is reached, the switch terminates */
}
```

"C:\Users\Anuradha\Desktop\C Programming\July25\bin\Debug\July25.exe"

```
Enter the first number: 5
Enter the second number: 3

Enter
  + for Addition
  - for Subtraction
  * for Multiplication
  / for Division : /

Division:
5.000 / 3.000 = 1.667

Process returned 0 (0x0)   execution time : 11.778 s
Press any key to continue.
```

3)

```c
#include <stdio.h>
int main()
{
    char x;
    printf("\nEnter \n  'C' for Circumference of Circle \n  'A' for Area of Circle"
           "\n  'V' for Volume of Sphere : ");
    scanf(" %c",&x);
    //user chooses a preferred calculation among circumference, area and volume

    float r;
    printf("\n\nEnter the Radius value(m) : ");
    scanf("%f",&r);
    //user inputs the radius value which the program reads in

    printf("\n");
    switch(x)
    {
        case 'C' :printf("Circumference of Circle: %.2f\n",2*3.14*r); break;
        case 'A' :printf("Area of Circle: %.2f\n",3.14*r*r); break;
        case 'V' :printf("Volume of Sphere: %.2f\n",(4/3)*3.14*r*r*r); break;
        default  :printf("Invalid Request\n");
    }
    //according to the user-input, the preferred calculation is performed and
    //printed on the console
    //switch is a control statement which allows a value to change control of execution
}
```

"C:\Users\Anuradha\Desktop\C Programming\July26\bin\Debug\July26.exe"

```
Enter
  'C' for Circumference of Circle
  'A' for Area of Circle
  'V' for Volume of Sphere : C


Enter the Radius value(m) : 7

Circumference of Circle: 43.96

Process returned 0 (0x0)   execution time : 29.571 s
Press any key to continue.
```

4)

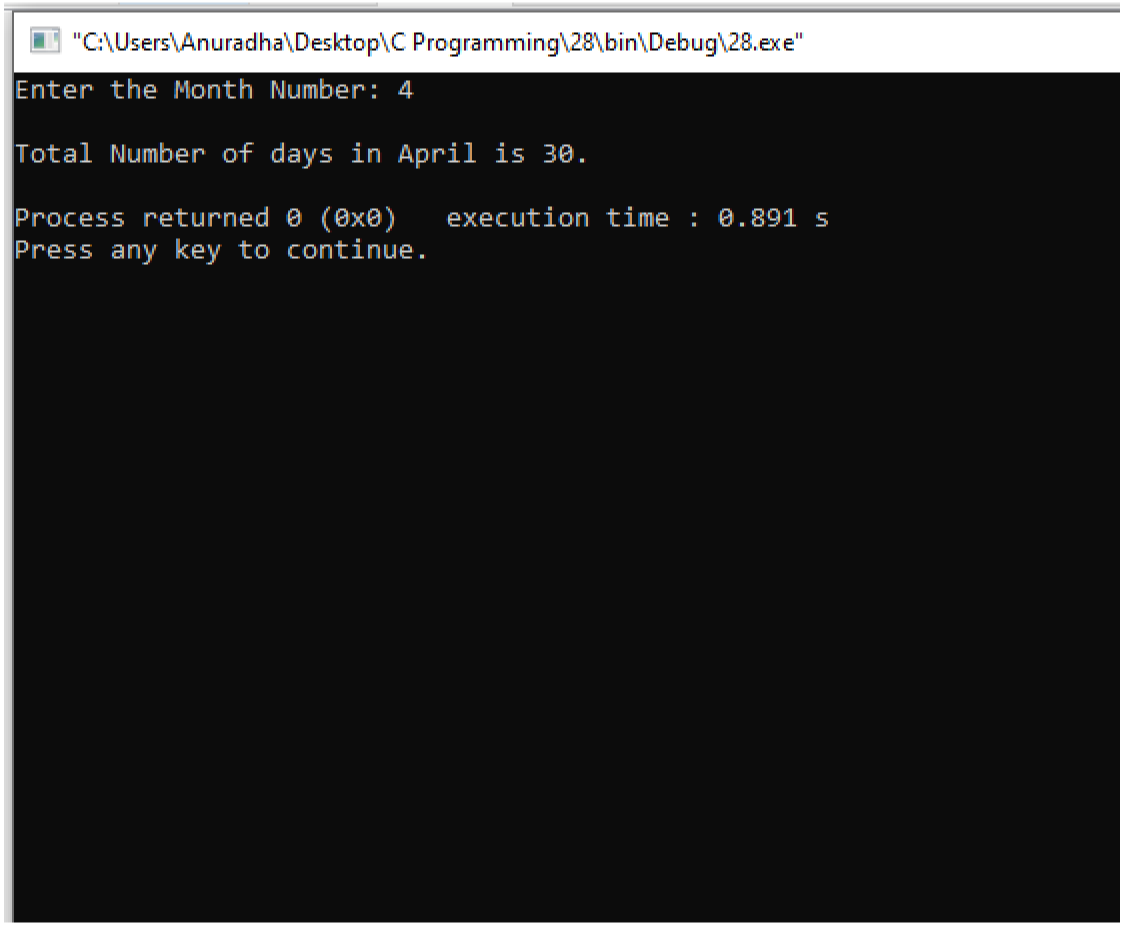
```
#include <stdio.h>
int main()
{
    char v;
    printf("Enter a character : ");
    scanf("%c",&v);
    //program reads in the user-input character

    switch(v)
    {
        case 'a':printf("%c is a vowel.\n",v); break;
        case 'e':printf("%c is a vowel.\n",v); break;
        case 'i':printf("%c is a vowel.\n",v); break;
        case 'o':printf("%c is a vowel.\n",v); break;
        case 'u':printf("%c is a vowel.\n",v); break;

        case 'A':printf("%c is a vowel.\n",v); break;
        case 'E':printf("%c is a vowel.\n",v); break;
        case 'I':printf("%c is a vowel.\n",v); break;
        case 'O':printf("%c is a vowel.\n",v); break;
        case 'U':printf("%c is a vowel.\n",v); break;

        default :printf("%c is not a vowel.\n",v);
    }
    /* the user-input character is tested for equality
    against the list of 5 vowels */
}
```

"C:\Users\Anuradha\Desktop\C Programming\27\bin\Debug\27.exe"

```
Enter a character : u
u is a vowel.

Process returned 0 (0x0)   execution time : 6.468 s
Press any key to continue.
```

5)

```c
#include <stdio.h>
int main()
{
    int m;
    printf("Enter the Month Number: ");
    scanf("%d",&m);
    //program reads in the user-entered month number

    printf("\n");
    switch(m)
    {
        case 1:printf("Total Number of days in January is 31.\n"); break;
        case 2:printf("Total Number of days in February is 28.\n"); break;
        case 3:printf("Total Number of days in March is 31.\n"); break;
        case 4:printf("Total Number of days in April is 30.\n"); break;
        case 5:printf("Total Number of days in May is 31.\n"); break;
        case 6:printf("Total Number of days in June is 30.\n"); break;
        case 7:printf("Total Number of days in July is 31.\n"); break;
        case 8:printf("Total Number of days in August is 31.\n"); break;
        case 9:printf("Total Number of days in September is 30.\n"); break;
        case 10:printf("Total Number of days in October is 31.\n"); break;
        case 11:printf("Total Number of days in November is 30.\n"); break;
        case 12:printf("Total Number of days in December is 31.\n"); break;
    }
    //case labels in the switch block are month numbers.
    //switch executes the statement that follows the case label
    //that matches the user-input month number
}
```


"C:\Users\Anuradha\Desktop\C Programming\28\bin\Debug\28.exe"
Enter the Month Number: 4
Total Number of days in April is 30.
Process returned 0 (0x0)   execution time : 0.891 s
Press any key to continue.

Part B

Section A

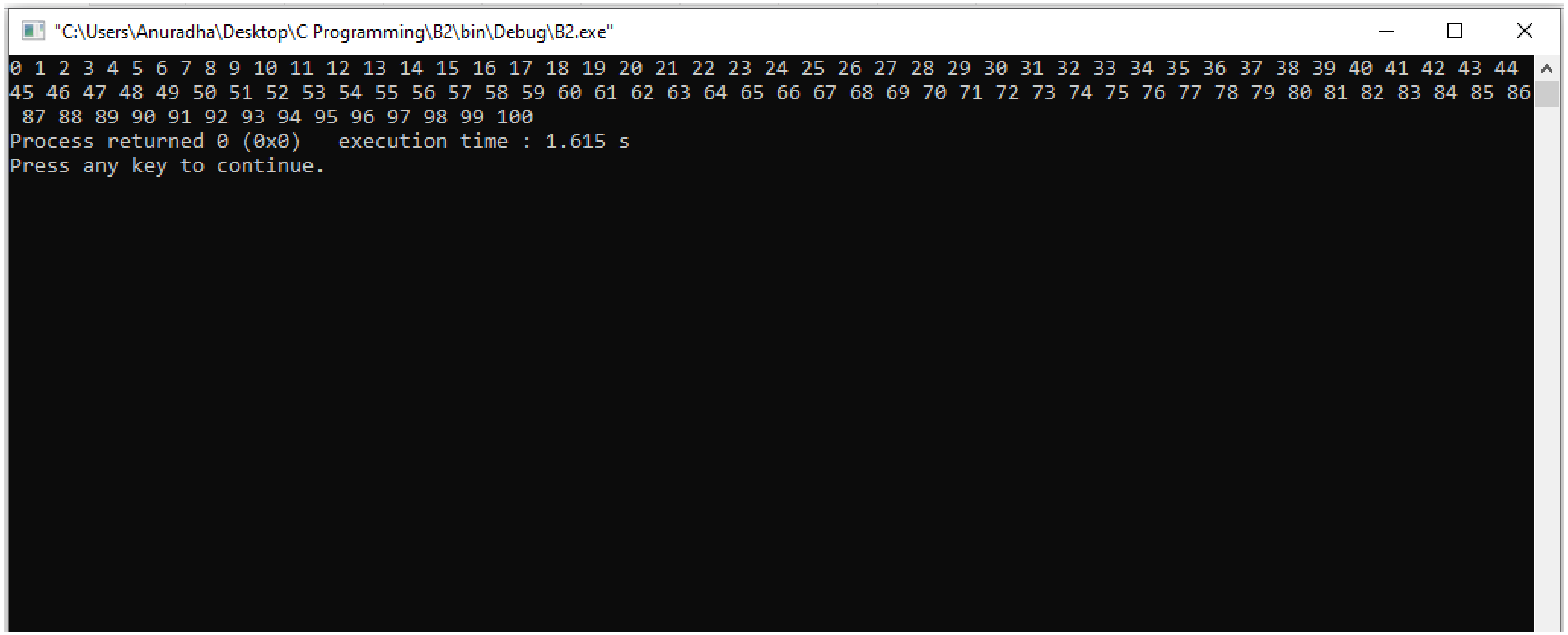
1)

```
#include <stdio.h>
int main()
{
    int x=0;
    while(x<=100)
    {
        printf("%d ",x);
        x++;
    }

    /* loop will iterate till the condition
    remains true.

    Initially 0 is assigned to x.

    Loop will terminate when x is not less than
    or not equal to 100.

    x value will be printed during each iteration. */
}
```

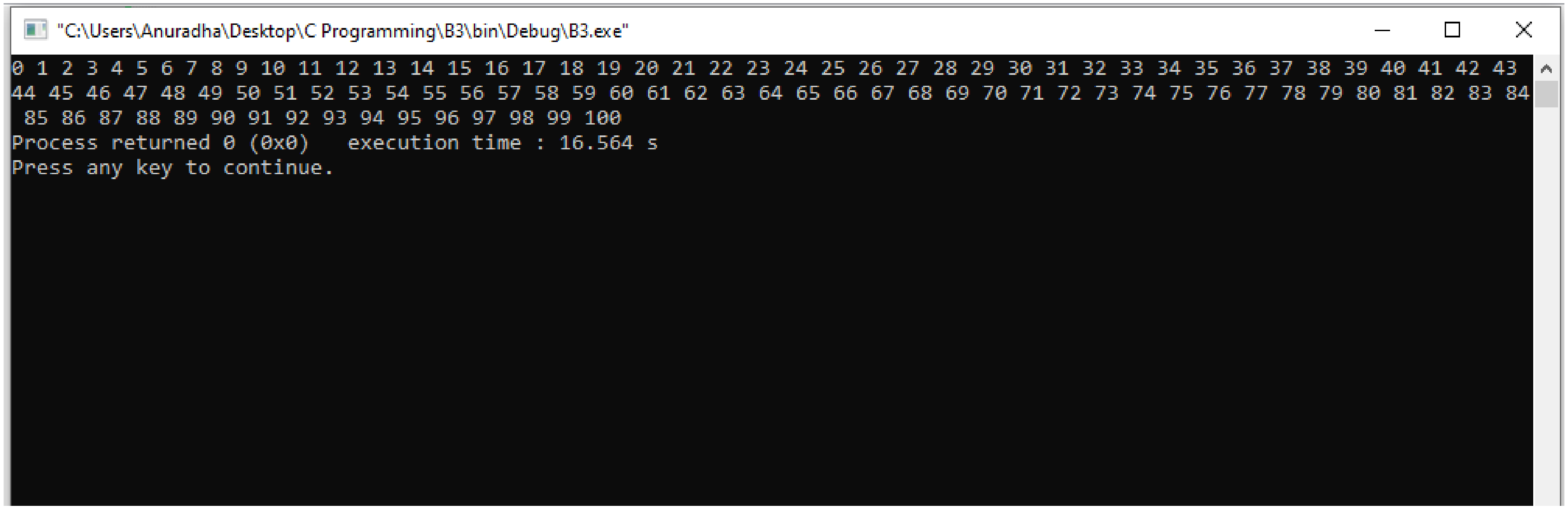
"C:\Users\Anuradha\Desktop\C Programming\July26\bin\Debug\July26.exe"

```
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44
45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86
 87 88 89 90 91 92 93 94 95 96 97 98 99 100
Process returned 0 (0x0)   execution time : 15.847 s
Press any key to continue.
```

```c
#include <stdio.h>
int main()
{
    int x=0;
    do{
        printf("%d ",x);
        x++;
    }while(x<=100);

    /*do while loop is similar to the while loop
    with one exception that it executes the statements
    inside the body of do-while before checking the
    condition.

    On the other hand in the while loop, first the
    condition is checked and then the statements
    in while loop are executed.

    Even if a condition is false at the first place,
    the do while would run once. */

}
```


"C:\Users\Anuradha\Desktop\C Programming\B2\bin\Debug\B2.exe"
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44
45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86
87 88 89 90 91 92 93 94 95 96 97 98 99 100
Process returned 0 (0x0)   execution time : 1.615 s
Press any key to continue.

```c
#include <stdio.h>
int main()
{
    for(int x=0;x<=100;x++)
    {
        printf("%d ",x);
    }

    /*
    In for loop, a loop variable is used to control the loop.
    First initialize this loop variable to some value,
    then check whether this variable is less than or
    greater than counter value.

    If statement is true, then loop body is executed and loop
    variable gets updated.

    Steps are repeated till exist condition comes. */
}
```


"C:\Users\Anuradha\Desktop\C Programming\B3\bin\Debug\B3.exe"
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43
44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
Process returned 0 (0x0)   execution time : 16.564 s
Press any key to continue.

2)

```c
#include <stdio.h>
int main()
{
    int marks,total;
    for(int x=0;x<10;x++)
    {
        printf("Enter Mark %02d: ",x+1);
        scanf("%d",&marks);
        total+=marks;
    }
    //each time, the user enters a new mark, the total
    //gets updated (the new marks gets added to the total)

    printf("\nTotal: %d\n",total);
    //total is displyed

    float avg=(float)total/10;
    //average is converted to float as total was stored
    //in an integer variable

    printf("Average: %.2f\n",avg);
```

```c

    if(avg<50)
        printf("Fail\n");
    else
        printf("Pass\n");
    //if the average is greater than 50, "Pass" will be printed
    //if not, "Fail" will be displayed on the console
}

```

"C:\Users\Anuradha\Desktop\C Programming\B4\bin\Debug\B4.exe"

```
Enter Mark 01: 40
Enter Mark 02: 59
Enter Mark 03: 59
Enter Mark 04: 49
Enter Mark 05: 69
Enter Mark 06: 68
Enter Mark 07: 96
Enter Mark 08: 94
Enter Mark 09: 64
Enter Mark 10: 76

Total: 674
Average: 67.40
Pass

Process returned 0 (0x0)   execution time : 22.527 s
Press any key to continue.
```

3)

```c
#include <stdio.h>
int main()
{
    int num;
    printf("Enter a number: ");
    scanf("%d",&num);

    int x=1,f=1;
    while(x<=num)
    {
        f=f*x;
        x++;
    }
    /* If user inputs 6, then the factorial of 6
    will be calculated as 6*5*4*3*2*1 = 720 */

    printf("\nFactorial of %d: \n%d\n",num,f);
}

```

"C:\Users\Anuradha\Desktop\C Programming\B5\bin\Debug\B5.exe"

```
Enter a number: 5

Factorial of 5:
120

Process returned 0 (0x0)   execution time : 8.992 s
Press any key to continue.
```

4)

```c
#include <stdio.h>
int main()
{
    int num,sum=0;
    printf("Enter a number: ");
    scanf("%d",&num);

    while(num!=0)
    {
        sum=sum+num%10;
        num=num/10;
    }

    /*the while loop will iterate till
    num becomes equal to 0.

    "num%10" is used to isolate the last
    digit of num

    "num/10" is used to get rid of the last
    digit of num */

    printf("\nSum: %d\n",sum);
}

```

"C:\Users\Anuradha\Desktop\C Programming\B6\bin\Debug\B6.exe"

```
Enter a number: 123

Sum: 6

Process returned 0 (0x0)   execution time : 11.462 s
Press any key to continue.
```

5)

```c
#include <stdio.h>
int main()
{
    int num;
    printf("Enter a number: ");
    scanf("%d",&num);
    int x=num;

    int rev=0,a=0;
    do{
        a=num%10;
        rev=rev*10+a;
        num=num/10;

    }while(num>0);

    printf("\nReverse Order of %d:\n%d\n",x,rev);

    //modulus operator(%) is used to obtain the digits
    //of the number

    //the loop will iterate once without checking the
    //condition and the iteration will continue till
    //the value of num is greater than 0
}

```

"C:\Users\Anuradha\Desktop\C Programming\B7\bin\Debug\B7.exe"

```
Enter a number: 123

Reverse Order of 123:
321

Process returned 0 (0x0)   execution time : 3.589 s
Press any key to continue.
```

6)

```
#include <stdio.h>
int main()
{
    int b;
    printf("Enter the Base Value: ");
    scanf("%d",&b);

    int e;
    printf("Enter the Exponent: ");
    scanf("%d",&e);

    int ans=1;

    for(int x=1;x<=e;x++)
    {
        ans=ans*b;
    }

    printf("\n%d to the power %d is %d.\n",b,e,ans);
```

```
    /*The program takes two integers from the user
    (a base number and an exponent) and calculates the power.

    If the base number is 2 and the power is 3,
    the answer is equal to 2*2*2.

    the above program works only if the exponent is
    a positive integer. */
}

```

"C:\Users\Anuradha\Desktop\C Programming\B8\bin\Debug\B8.exe"

```
Enter the Base Value: 2
Enter the Exponent: 3

2 to the power 3 is 8.

Process returned 0 (0x0)   execution time : 6.304 s
Press any key to continue.
```

7)

```
#include <stdio.h>
int main()
{
    int f1=0,f2=1,f3;

    printf("First 10 Fibonacci Numbers:\n\n");
    printf("%d %d ",f1,f2);
    //first two Fibonacci numbers are printed

    for(int x=3;x<=10;x++)
    {
        f3=f1+f2;
        printf("%d ",f3);
        f1=f2;
        f2=f3;
    }

    //All other terms are obtained by adding the preceding two terms
    //Terms are updated inside the for loop
    //Before entering the for loop, program checks if the number of
    //terms is valid

    printf("\n\n");

}
```

"C:\Users\Anuradha\Desktop\C Programming\B9\bin\Debug\B9.exe"

```
First 10 Fibonacci Numbers:

0 1 1 2 3 5 8 13 21 34


Process returned 0 (0x0)   execution time : 7.421 s
Press any key to continue.
```

8)

```c
#include <stdio.h>
int main()
{
    int num,copy_of_num,sum=0,rem;

    printf("Enter a number: ");
    scanf("%d",&num);
    //user-input is stored in variable num

    copy_of_num = num;
    /*value of variable num would change in the
      below while loop so we are storing it in
      another variable to compare the results
      at the end of program */

    while (num!=0)
    {
        rem=num%10;
        sum=sum+(rem*rem*rem);
        num=num/10;
    }

    /*we are adding cubes of every digit and
      storing the sum in variable sum */

```

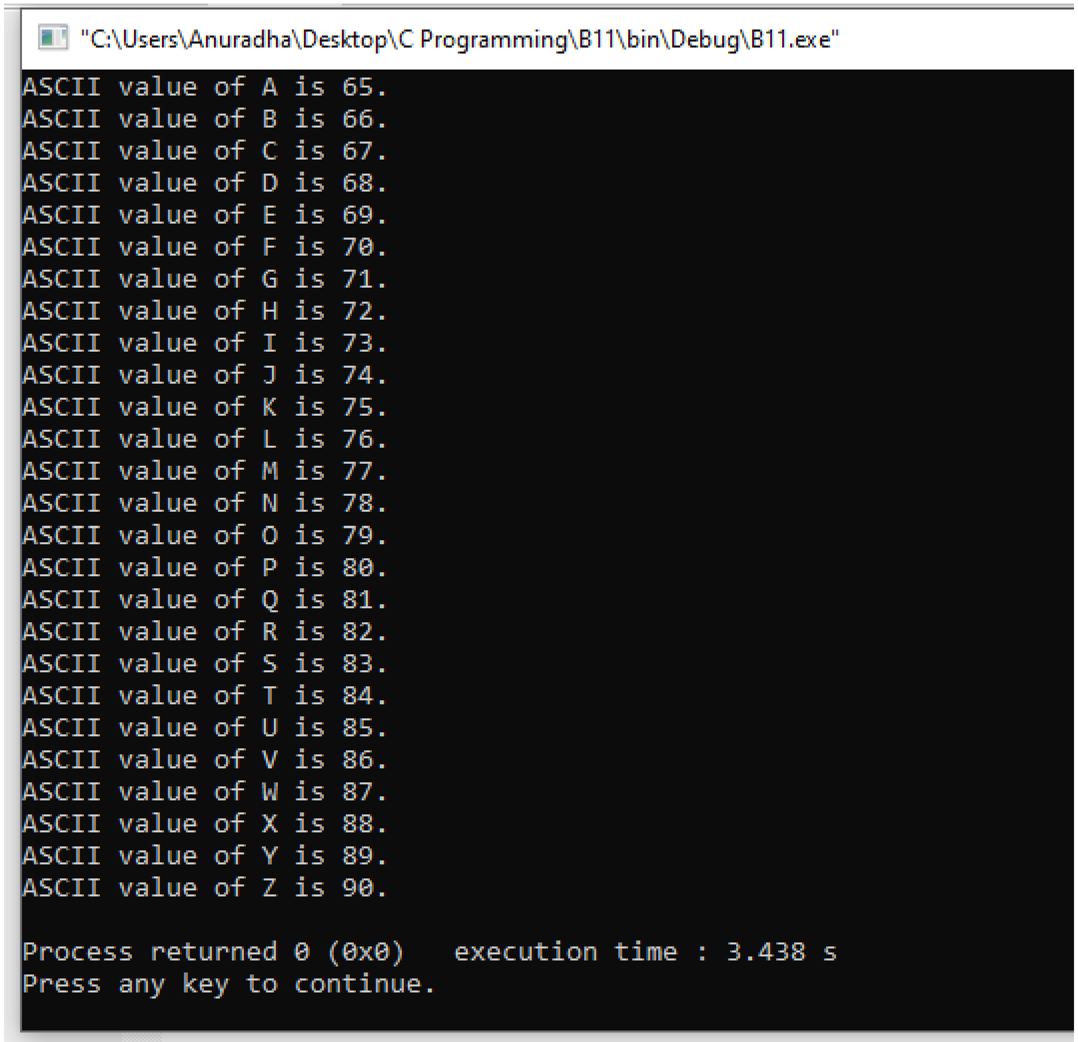
```c
    if(copy_of_num == sum)
        printf("\n%d is an Armstrong Number.\n",copy_of_num);
    else
        printf("\n%d is not an Armstrong Number.\n",copy_of_num);

    /*If sum of cubes of every digit is equal to
      number itself, then the number is Armstrong */
}

```

"C:\Users\Anuradha\Desktop\C Programming\B10\bin\Debug\B10.exe"

```
Enter a number: 407

407 is an Armstrong Number.

Process returned 0 (0x0)   execution time : 2.279 s
Press any key to continue.
```

9)

```c
#include <stdio.h>
int main()
{
    for(int x=65;x<=90;x++)
    {
        printf("ASCII value of %c is %d.\n",x,x);
    }

    /*
    %c is the format string for a single character.
    %d is the conversion specifier for an integer value.
    By casting the char to an integer,
    you will get the ASCII value */
}
```



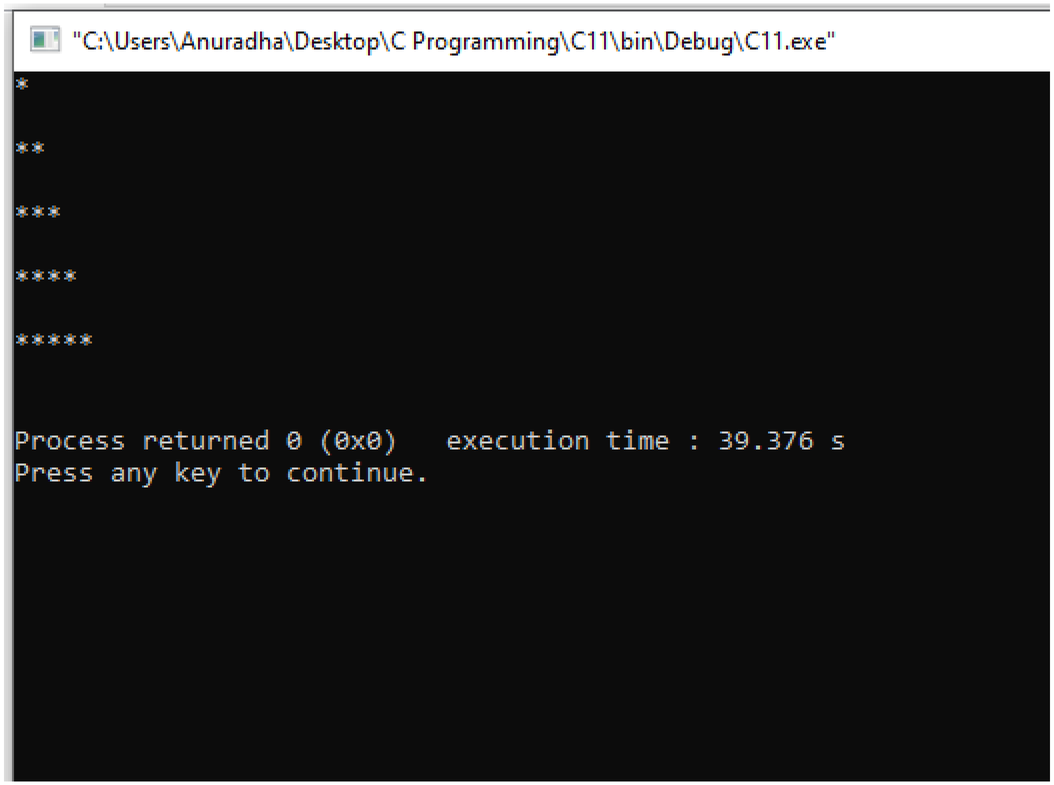
"C:\Users\Anuradha\Desktop\C Programming\B11\bin\Debug\B11.exe"

```
ASCII value of A is 65.
ASCII value of B is 66.
ASCII value of C is 67.
ASCII value of D is 68.
ASCII value of E is 69.
ASCII value of F is 70.
ASCII value of G is 71.
ASCII value of H is 72.
ASCII value of I is 73.
ASCII value of J is 74.
ASCII value of K is 75.
ASCII value of L is 76.
ASCII value of M is 77.
ASCII value of N is 78.
ASCII value of O is 79.
ASCII value of P is 80.
ASCII value of Q is 81.
ASCII value of R is 82.
ASCII value of S is 83.
ASCII value of T is 84.
ASCII value of U is 85.
ASCII value of V is 86.
ASCII value of W is 87.
ASCII value of X is 88.
ASCII value of Y is 89.
ASCII value of Z is 90.

Process returned 0 (0x0)   execution time : 3.438 s
Press any key to continue.
```

10)

```c
#include <stdio.h>
int main()
{
    for(int x=1;x<=5;x++)
    {
        for(int y=1;y<=x;y++)
        {
            printf("*");
        }
        printf("\n\n");
    }

    /*Nested for loop is used in this program.

    Nesting of loops is a feature in C that allows the
    looping of statements inside another loop.

    Any number of loops can be defined inside another
    loop.

    1 is initialized to x. Then program control passes to
    x<=5 and checks whether the condition is true or false.

    If the condition is true, the program control passes to
    the inner loop.
```

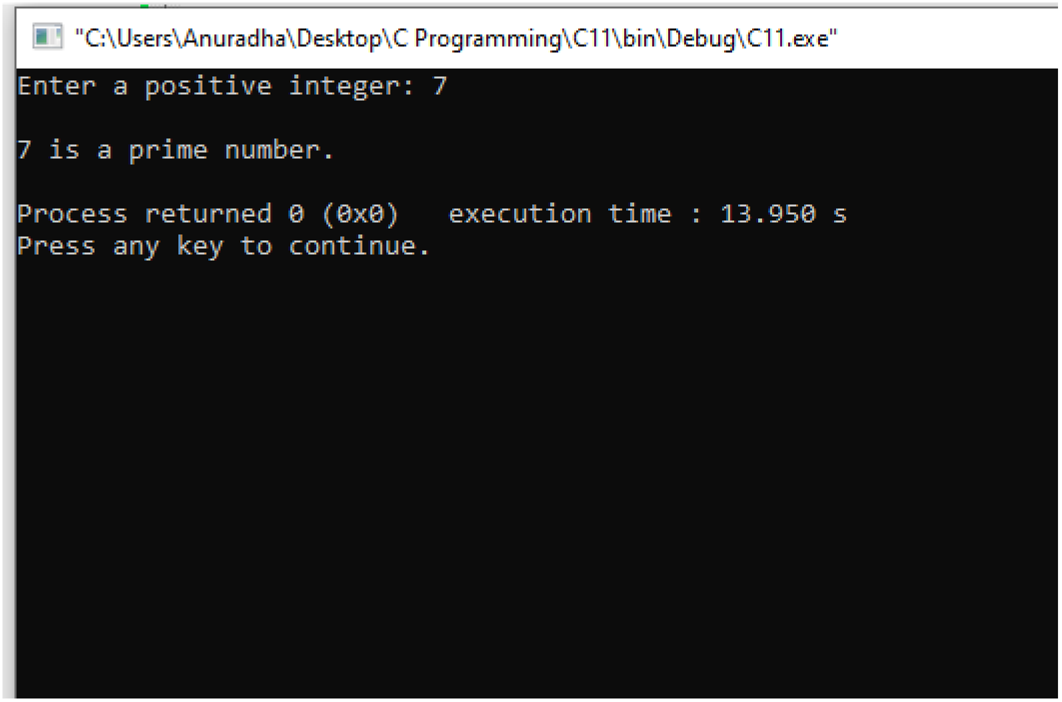
```
    The inner loop will get executed until the condition
    is true.

    After the execution of the inner loop, the control
    moves back to the update of the outer loop. (x++)

    After the increment, the condition is checked again.
    (x<=5)

    If the condition is true, then the inner loop will
    be executed again.

    This process will continue until the condition of the
    outer loop is true. */
}

```


"C:\Users\Anuradha\Desktop\C Programming\C11\bin\Debug\C11.exe"
*
**
***
****
*****
Process returned 0 (0x0)   execution time : 39.376 s
Press any key to continue.

11)

```c
#include <stdio.h>
int main()
{
    int n, i, flag = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    for (i = 2; i <= n / 2; ++i) {

        // condition for non-prime
        if (n % i == 0) {
            flag = 1;
            break;
        }
    }

    if (n == 1) {
        printf("\n1 is neither prime nor composite.\n");
    }
    else {
        if (flag == 0)
            printf("\n%d is a prime number.\n", n);
        else
            printf("\n%d is not a prime number.\n", n);
    }
```

```
    /*A prime number is a positive number which is divisible only by 1
    and itself. (ex. 2,3,5,7,11,13,17)

    In the above program, a for loop is iterated from i=2 to i<n/2.

    In each iteration, whether n is perfectly divisible by i is checked using

    if(n%i==0){
    }

    If n is perfectly divisible by i, n is not a prime number. In this case, flag
    is set to 1, and the loop is terminated using the break statement.

    After the loop, if n is a prime number, flag will still be 0.
    However, if n is a non-prime number, flag will be 1.

    */

}

```


"C:\Users\Anuradha\Desktop\C Programming\C11\bin\Debug\C11.exe"
Enter a positive integer: 7
7 is a prime number.
Process returned 0 (0x0)   execution time : 13.950 s
Press any key to continue.

12)

```c
#include <stdio.h>

int main()
{
    int i, num;

    /* Input number from user */
    printf("Enter a number to find its factor: ");
    scanf("%d", &num);

    printf("\nAll factors of %d are: \n", num);

    /* Iterate from 1 to num */
    for(i=1; i<=num; i++)
    {
        /*
         * If num is exactly divisible by i
         * Then i is a factor of num
         */
        if(num % i == 0)
        {
            printf("%d, ",i);
        }
    }

}
```

"C:\Users\Anuradha\Desktop\C Programming\C11\bin\Debug\C11.exe"

```
Enter a number to find its factor: 50

All factors of 50 are:
1, 2, 5, 10, 25, 50,
Process returned 0 (0x0)   execution time : 2.765 s
Press any key to continue.
```

12) – Question No. 12 was repeated int the Lab Sheet

```
#include <stdio.h>
int main()
{
    int x=1,num,sum=0;

    do
    {
        printf("Enter a number:  ");
        scanf("%d",&num);

        sum+=num;
        x++;
    }while(num!=-1);

    printf("\nSum of all entered numbers: %d\n",sum+1);
    //Sum is calculated disregarding -1

    /*
    The initial value of sum is 0 and it will get updated
    each time user inputs a new value.

    The loop will iterate till the user inputs -1 to the
    program. (num!=-1)

    */
}
```

"C:\Users\Anuradha\Desktop\C Programming\C11\bin\Debug\C11.exe"

```
Enter a number: 3
Enter a number: 2
Enter a number: 3
Enter a number: 4
Enter a number: -1

Sum of all entered numbers: 12

Process returned 0 (0x0)   execution time : 8.139 s
Press any key to continue.
```

13)

```c
#include <stdio.h>
int main()
{
    int arr[10];
    for(int x=0;x<10;x++)
    {
        printf("Enter a value for arr[%d]: ",x);
        scanf("%d",&arr[x]);
        //user inputs values for each array element and the
        //program stores them in each array element
    }

    printf("\n\nArray:\n");

    for(int x=0;x<10;x++)
    {
        printf("%d\t",arr[x]);
        //values in each array element is printed out
    }

    printf("\n");

    //array is a collection of data items, all of the same data type
    //In this program, single-dimensional arrays are being used
}
```

"C:\Users\Anuradha\Desktop\C Programming\C11\bin\Debug\C11.exe"

```
Enter a value for arr[0]: 3
Enter a value for arr[1]: 2
Enter a value for arr[2]: 6
Enter a value for arr[3]: 4
Enter a value for arr[4]: 8
Enter a value for arr[5]: 9
Enter a value for arr[6]: 9
Enter a value for arr[7]: 6
Enter a value for arr[8]: 7
Enter a value for arr[9]: 9


Array:
3       2       6       4       8       9       9       6       7       9

Process returned 0 (0x0)   execution time : 9.623 s
Press any key to continue.
```

14)

```c
#include <stdio.h>
int main()
{
    int arr[10],ec=0;
    for(int x=0;x<10;x++)
    {
        printf("Enter a value for arr[%d]: ",x);
        scanf("%d",&arr[x]);

        if(arr[x]%2==0)
            ec++;
        //each time the user inputs a value to an array element, it is divided
        //by 2, and if the remainder is 0 (if the modulus is 0),
        //that user-input is counted in by the variable ec
    }

    printf("\n\nArray:\n");

    for(int x=0;x<10;x++)
    {
        printf("%d\t",arr[x]);
    }

    printf("\n");
    printf("\nTotal Even Count: %d\n",ec);
    //the total number of even numbers, entered by the user is displayed
}
```

"C:\Users\Anuradha\Desktop\C Programming\Even\bin\Debug\Even.exe"

```
Enter a value for arr[0]: 2
Enter a value for arr[1]: 2
Enter a value for arr[2]: 2
Enter a value for arr[3]: 2
Enter a value for arr[4]: 2
Enter a value for arr[5]: 2
Enter a value for arr[6]: 2
Enter a value for arr[7]: 2
Enter a value for arr[8]: 2
Enter a value for arr[9]: 2


Array:
2       2       2       2       2       2       2       2       2       2

Total Even Count: 10

Process returned 0 (0x0)   execution time : 11.435 s
Press any key to continue.
```

Section B

1)

```c
#include <stdio.h>
int main()
{
    int x=1,no,p=0,n=0,z=0;

    while(x<=10)
    {
        printf("Enter Number %02d: ",x);
        scanf("%d",&no);

        if(no>0)
            p++;
        //if the user-input is greater than 0, that is a positive number
        else if(no<0)
            n++;
        //if the user-input is less than 0, that is a negative number
        else
            z++;
        //if any user-input is neither greater or less than 0, that number is 0

        x++;
        //counter is incremented for the next round
    }
```

```c
    printf("\n\n");
    printf("Total count of Positives: %d\n",p);
    printf("Total count of Negatives: %d\n",n);
    printf("Total count of Zeros: %d\n",z);
    //Total Positive, Negative, Zero counts are displayed
}
```

"C:\Users\Anuradha\Desktop\C Programming\Even\bin\Debug\Even.exe"

```
Enter Number 01: 1
Enter Number 02: 2
Enter Number 03: 3
Enter Number 04: -1
Enter Number 05: -2
Enter Number 06: -3
Enter Number 07: 0
Enter Number 08: 0
Enter Number 09: 0
Enter Number 10: 0


Total count of Positives: 3
Total count of Negatives: 3
Total count of Zeros: 4

Process returned 0 (0x0)   execution time : 16.496 s
Press any key to continue.
```

2)

```c
#include <stdio.h>
int main()
{
    int marks[10],sum=0;
    for(int x=0;x<10;x++)
    {
        printf("Enter Mark %02d: ",x+1);
        scanf("%d",&marks[x]);
        sum=sum+marks[x];
    }

    float avg=(float)sum/10;
    //float() converts integer to floating-point values
    printf("\nAverage: %.2f",avg);

    int min=marks[0];
    /*Initially first mark is assigned to min
    Loop checks every other mark to see if it is less than the first one */
    for(int x=1;x<10;x++)
    {
        if(marks[x]<min)
            min=marks[x];
    }
    printf("\nMinimum Marks: %d\n",min);
```

```c
    int max=marks[0];
    /*Initially first mark is assigned to max
    Loop checks every other mark to see if it is greater than the first one */
    for(int x=1;x<10;x++)
    {
        if(marks[x]>max)
            max=marks[x];
    }
    printf("Maximum Marks: %d\n",max);

}
```

"C:\Users\Anuradha\Desktop\C Programming\Even\bin\Debug\Even.exe"

```
Enter Mark 01: 53
Enter Mark 02: 75
Enter Mark 03: 77
Enter Mark 04: 45
Enter Mark 05: 75
Enter Mark 06: 98
Enter Mark 07: 99
Enter Mark 08: 76
Enter Mark 09: 86
Enter Mark 10: 37

Average: 72.10
Minimum Marks: 37
Maximum Marks: 99

Process returned 0 (0x0)   execution time : 12.950 s
Press any key to continue.
```

3)

```
#include <stdio.h>
int main()
{
    float price[10],sum=0;
    int y=0;

    for(int x=0;x<10;x++)
    {
        printf("Enter Price of Item No. %02d: Rs.",x+1);
        scanf("%f",&price[x]);
        sum=sum+price[x];
        //sum of all items are calculated

        if(price[x]>200)
        y++;
        //if an item price is more than Rs.200, the condition is true
        //and y will be incremented by 1

    }

    float avg=sum/10;
    //average price of an item is calculated

    printf("\nAverage Value of an Item: Rs.%.2f",avg);
    printf("\nNumber of items greater than Rs.200/= : %d\n\n",y);
    //Average value and item count are displayed
}

```

```
"C:\Users\Anuradha\Desktop\C Programming\Even\bin\Debug\Even.exe"
Enter Price of Item No. 01: Rs.100
Enter Price of Item No. 02: Rs.200
Enter Price of Item No. 03: Rs.300
Enter Price of Item No. 04: Rs.400
Enter Price of Item No. 05: Rs.500
Enter Price of Item No. 06: Rs.600
Enter Price of Item No. 07: Rs.700
Enter Price of Item No. 08: Rs.800
Enter Price of Item No. 09: Rs.900
Enter Price of Item No. 10: Rs.900

Average Value of an Item: Rs.540.00
Number of items greater than Rs.200/= : 8


Process returned 0 (0x0)   execution time : 12.279 s
Press any key to continue.
```

4)

```c
#include <stdio.h>
int main()
{
    float bs;
    //variable to store the basic salary
    int x=1,en,c=0;
    do{
        printf("Enter Employee Number: ");
        scanf("%d",&en);
        //program reads in the Employee Number of each employee
        printf("Enter Basic Salary: Rs.");
        scanf("%f",&bs);
        //program reads in the Basic Salary of each employee

        if(bs>=5000)
            c++;
        //if the Basic Salary of an employee is >=5000, c gets incremented

        x++;
        //counter increments by 1 for the loop to iterate again

    }while(en!=-999);
    //loop will iterate till the entered Employee Number is -999

    printf("Number of Employees whose Basic Salary is more than or equal to Rs.5000/=: %d\n\n",c);
    //No of employee earning Rs.5000 or more are printed on the console
}
```

"C:\Users\Anuradha\Desktop\C Programming\July28\bin\Debug\July28.exe"

```
Enter Employee Number: 111
Enter Basic Salary: Rs.3000
Enter Employee Number: 222
Enter Basic Salary: Rs.5000
Enter Employee Number: 333
Enter Basic Salary: Rs.6000
Enter Employee Number: 444
Enter Basic Salary: Rs.9000
Enter Employee Number: 555
Enter Basic Salary: Rs.8000
Enter Employee Number: 666
Enter Basic Salary: Rs.6000
Enter Employee Number: 777
Enter Basic Salary: Rs.9000
Enter Employee Number: -999
Enter Basic Salary: Rs.-
Number of Employees whose Basic Salary is more than or equal to Rs.5000/=: 7


Process returned 0 (0x0)   execution time : 34.020 s
Press any key to continue.
```

Practical 05

1)

```
#include <stdio.h>
int main()
{
    int arr[10],min,max,sum=0;
    float avg;

    for(int x=0;x<10;x++)
    {
        printf("Enter a value for arr[%d]: ",x);
        scanf("%d",&arr[x]);
        //user inputs values to 10 array elements
    }

    max=arr[0];
    //first array element is assigned to min
    min=arr[0];
    //first array element is assigned to max

```

```c
    for(int x=1;x<10;x++)
    {
        if(arr[x]>max)
            max=arr[x];

        if(arr[x]<min)
        min=arr[x];

        sum=sum+arr[x];
        //sum gets updated with each iteration
    }

    printf("\nMinimum Value: %d\n",min);
    printf("Maximum Value: %d\n",max);
    printf("Average Value: %.2f\n",(float)sum/10);
    printf("Reverse Order of Values:\n");

    for(int x=9;x>=0;x--)
    {
        printf("%d ",arr[x]);
        //the for loop iterates till x>=0 condition is satisfied
        //x is decremented by 1 after each round
    }
    printf("\n\n");

}
```

"C:\Users\Anuradha\Desktop\All C\Week 10\C Workshop\Question 7\bin\Debug\Question 7.exe"

```
Enter a value for arr[0]: 0
Enter a value for arr[1]: 1
Enter a value for arr[2]: 2
Enter a value for arr[3]: 3
Enter a value for arr[4]: 4
Enter a value for arr[5]: 5
Enter a value for arr[6]: 6
Enter a value for arr[7]: 7
Enter a value for arr[8]: 8
Enter a value for arr[9]: 9

Minimum Value: 0
Maximum Value: 9
Average Value: 4.50
Reverse Order of Values:
9 8 7 6 5 4 3 2 1 0


Process returned 0 (0x0)   execution time : 12.404 s
Press any key to continue.
```

2)

```c
#include <stdio.h>
int main()
{
    int s;
    printf("What is the size of the array? ");
    scanf("%d",&s);
    //user determines the no. of elements in the array

    int arrA[s],arrB[s],arrC[s],arrD[s],ssA=0,ssB=0,sP=0;
    for(int x=0;x<s;x++)
    {
        printf("\nEnter a value for arrA[%d]: ",x);
        scanf("%d",&arrA[x]);
        ssA+=arrA[x];
        //scalar sum of arrA is calculated by adding the values of
        //each element of the arrA

        printf("Enter a value for arrB[%d]: ",x);
        scanf("%d",&arrB[x]);
        ssB+=arrB[x];
        //scalar sum of arrB is calculated by adding the values of
        //each element of the arrB
```

```c

        arrC[x]=arrA[x]+arrB[x];
        //values in each relative element in arrA and arrB are added
        //and stored in arrC to print the Vector Sum
        arrD[x]=arrA[x]*arrB[x];
        //values in relative elements in arrA and arrB are multiplied
        //and stored in arrD to print the Vector product
        sP+=arrD[x];
        //all the values in arrD are added together to get the Scalar Product
    }

    printf("\nScalar Sum of Array A: %d",ssA);
    printf("\nScalar Sum of Array B: %d",ssB);

    printf("\nVector Sum:\n");
    for(int x=0;x<s;x++)
    {
        printf("%d\t",arrC[x]);
    }

    printf("\nVector Product:\n");
    for(int x=0;x<s;x++)
    {
        printf("%d\t",arrD[x]);
    }
    printf("\nScalar Product: %d\n",sP);
}
```

"C:\Users\Anuradha\Desktop\All C\Week 10\C Workshop\Question 7\bin\Debug\Question 7.exe"

```
What is the size of the array? 3

Enter a value for arrA[0]: 1
Enter a value for arrB[0]: 1

Enter a value for arrA[1]: 2
Enter a value for arrB[1]: 2

Enter a value for arrA[2]: 3
Enter a value for arrB[2]: 3

Scalar Sum of Array A: 6
Scalar Sum of Array B: 6
Vector Sum:
2       4       6
Vector Product:
1       4       9
Scalar Product: 14

Process returned 0 (0x0)   execution time : 4.812 s
Press any key to continue.
```

Practical 06

1)

```c
#include <stdio.h>
int main()
{
    int A[3][3]; // Matrix 1
    int B[3][3]; // Matrix 2
    int C[3][3]; // Resultant matrix

    int row, col;

    /* Input elements in first matrix*/
    printf("Enter elements in matrix A of size 3x3: \n");
    for(row=0; row<3; row++)
    {
        for(col=0; col<3; col++)
        {
            scanf("%d", &A[row][col]);
        }
    }
    /* Input elements in second matrix */
    printf("\nEnter elements in matrix B of size 3x3: \n");
    for(row=0; row<3; row++)
    {
        for(col=0; col<3; col++)
        {
            scanf("%d", &B[row][col]);
        }
    }

```

```c

    /*
     * Add both matrices A and B entry wise or element wise
     * and stores result in matrix C
     */
    for(row=0; row<3; row++)
    {
        for(col=0; col<3; col++)
        {
            /* Cij = Aij + Bij */
            C[row][col] = A[row][col] + B[row][col];
        }
    }


    /* Print the value of resultant matrix C */
    printf("\nSum of matrices A+B = \n");
    for(row=0; row<3; row++)
    {
        for(col=0; col<3; col++)
        {
            printf("%d\t", C[row][col]);
        }
        printf("\n");
    }

}

```

"C:\Users\Anuradha\Desktop\All C\Week 10\C Workshop\Question 7\bin\Debug\Question 7.exe"

```
Enter elements in matrix A of size 3x3:
3 2 4
1 4 6
4 3 2

Enter elements in matrix B of size 3x3:
2 6 3
4 3 2
5 1 7

Sum of matrices A+B =
5       8       7
5       7       8
9       4       9

Process returned 0 (0x0)   execution time : 35.304 s
Press any key to continue.
```

Practical 07

1)

```c
#include <stdio.h>

 /* Declare function prototype that will send base and exponent values */
 float integerPower (float, int);
 int main (void)
 {

     float base;
     float answer;
     int exponent;

     /* Prompt the user to enter the base number */
     printf ("\nEnter the base number: ");
     scanf ("%f", &base);

     /* Prompt the user to enter the exponent */
     printf ("\nEnter the exponent: ");
     scanf ("%d", &exponent);

     /* Call function to calculate the value of the base to the exponent */
     answer = integerPower (base, exponent);

     /* Print the result */
     printf ("\nThe value of the exponential expression is: %.2f\n", answer);

     return 0;
  }

```

```
/* Description: Calculate the value of an exponential
** expression base ^ exponent
   Pre: base and exponent
   Post: returns value of exponential expression
*/
float integerPower (float base, int exponent)
{
    int i;
    float total = 1;

    for (i = 0; i < exponent; i++)
        total *= base;

    return total;
}
```

"C:\Users\Anuradha\Desktop\All C\Week 10\C Workshop\Question 7\bin\Debug\Question 7.exe"

```
Enter the base number: 2

Enter the exponent: 3

The value of the exponential expression is: 8.00

Process returned 0 (0x0)   execution time : 5.452 s
Press any key to continue.
```

2)

main.c main.c

```c
#include <stdio.h>
#include <math.h>

//function prototype
int seconds(int h, int m, int s);

int main()
{
    int hours; //current time's hours
    int minutes; //current time's minutes
    int secs; //current time's seconds
    int first; //first time, in seconds
    int second; //second time, in seconds
    int difference; //difference between two times, in seconds

    printf("%s", "Enter the first time as three integers: ");
    scanf("%d%d%d",&hours,&minutes,&secs);

    //calculate first time in seconds
    first=seconds(hours,minutes,secs);

    printf("%s","Enter the second time as three integers: ");
    scanf("%d%d%d",&hours,&minutes,&secs);

    //calculate second time in seconds
    second=seconds(hours,minutes,secs);

```

```c
    //calculate difference
    difference=fabs(first-second);

    //display difference
    printf("The difference between the times is %d seconds\n",difference);
} //end main

//seconds returns number of seconds since clock "struck 12"
//given input time as hours h, minutes m, seconds s
int seconds(int h, int m, int s)
{
    return 3600 * h + 60 * m + s;
}
```

"C:\Users\Anuradha\Desktop\All C\Week 10\C Workshop\Question 7\bin\Debug\Question 7.exe"

```
Enter the first time as three integers: 10 44 32
Enter the second time as three integers: 10 45 33
The difference between the times is 61 seconds

Process returned 0 (0x0)   execution time : 11.969 s
Press any key to continue.
```

3)

```
#include <stdio.h>
#include <conio.h>

void ftoc(int);
void ctof(int);

void main()
{
    int f,c=0;
    printf("Fahrenheit to Celsius : Celsius to Fahrenheit\n");
    //head of the chart
    for(f=32;f<=212;f++)
    {
        ftoc(f);
        if(c<=100)
            ctof(c);
        c++;
        printf("\n");
    }
    //the loop will iterate until the condition (f<=212) remains true
    getch();
    //function getch() is being called
}

```

```c
void ftoc(int f)
{
    float cf;
    cf=(f-32)*0.555555555;
    printf(" %d        %f",f,cf);
    /*prints a chart showing the Fahrenheit equivalents of all Celsius
    temperatures from 0 to 100 degrees */
}

void ctof(int c)
{
    float fc;
    fc=1.8*c+32;
    printf("        %d       %f",c,fc);
    /*prints a chart showing the Celsius equivalents of all Fahrenheit
    temperatures from 32 to 212 degrees

    outputs will be printed in a near tabular format*/
}
```

"C:\Users\Anuradha\Desktop\C Programming\July27\bin\Debug\July27.exe"

```
Fahrenheit to Celsius : Celsius to Fahrenheit
 32        0.000000        0       32.000000
 33        0.555556        1       33.799999
 34        1.111111        2       35.599998
 35        1.666667        3       37.400002
 36        2.222222        4       39.200001
 37        2.777778        5       41.000000
 38        3.333333        6       42.799999
 39        3.888889        7       44.599998
 40        4.444445        8       46.400002
 41        5.000000        9       48.200001
 42        5.555555        10       50.000000
 43        6.111111        11       51.799999
 44        6.666667        12       53.599998
 45        7.222222        13       55.400002
 46        7.777778        14       57.200001
 47        8.333333        15       59.000000
 48        8.888889        16       60.799999
 49        9.444445        17       62.599998
 50        10.000000        18       64.400002
 51        10.555555        19       66.199997
 52        11.111111        20       68.000000
 53        11.666667        21       69.800003
 54        12.222222        22       71.599998
 55        12.777778        23       73.400002
 56        13.333333        24       75.199997
 57        13.888889        25       77.000000
 58        14.444445        26       78.800003
 59        15.000000        27       80.599998
 60        15.555555        28       82.400002
 61        16.111111        29       84.199997
 62        16.666666        30       86.000000
 63        17.222221        31       87.800003
 64        17.777779        32       89.599998
 65        18.333334        33       91.400002
 66        18.888889        34       93.199997
 67        19.444445        35       95.000000
 68        20.000000        36       96.800003
 69        20.555555        37       98.599998
 70        21.111111        38       100.400002
 71        21.666666        39       102.199997
 72        22.222221        40       104.000000
 73        22.777779        41       105.800003
```

4)

```c
#include <stdio.h>
void smallest(float a,float b,float c)
//fnction header
//no return type, with parameters
{
    //below is the function body

    float min;
    //flaot varibale min is declared to store the samllest value
    min=a;
    //first user-input is assigned to min
    if(b<min)
        min=b;
    //if the second user-input is less than the first, second one is assigned to min
    if(c<min)
        min=c;
    //if the third user-input is less than the second, third one is assigned to min
    printf("\n\nSmallest float value: %.2f\n",min);
    //smallest flaot value is printed on the console
}
```

```
int main()
//main function where the smallest() function is being called
{
    float x,y,z;
    //3 float varibles are declared

    printf("Enter the first float value: ");
    scanf("%f",&x);
    //user inputs the first float value

    printf("Enter the second float value: ");
    scanf("%f",&y);
    //program reads in the second float value

    printf("Enter the third float value: ");
    scanf("%f",&z);
    //user inputs the third flaot value

    smallest(x,y,z);
    //smallest() function is called and values are passed
}

```

"C:\Users\Anuradha\Desktop\C Programming\final\bin\Debug\final.exe"

```
Enter the first float value: 6.4
Enter the second float value: 5.3
Enter the third float value: 9.5


Smallest float value: 5.30

Process returned 0 (0x0)   execution time : 6.959 s
Press any key to continue.
```